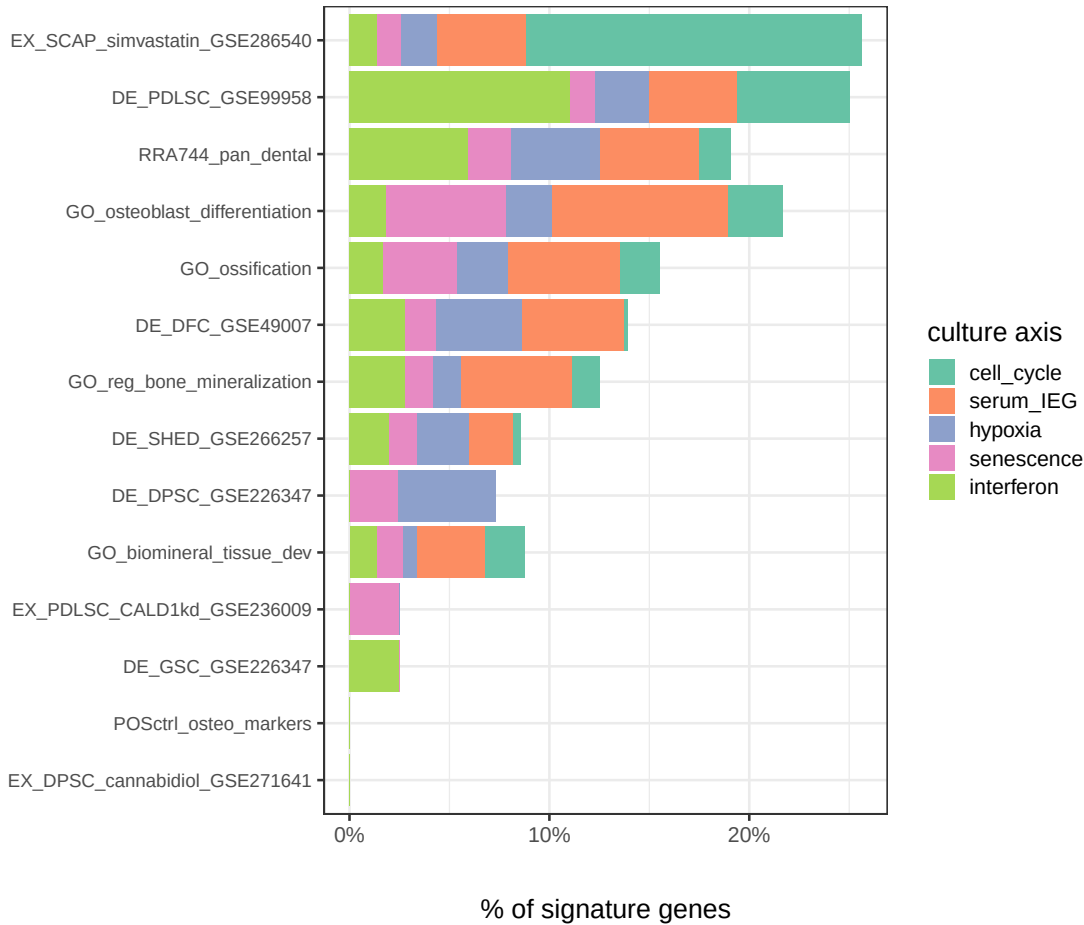


A**Culture-state fingerprint composition of each signature**

Fraction of signature genes belonging to in-vitro culture programs (cell-cycle, serum/immediate-early, hypoxia, senescence, interferon).

**B****Which culture programs are enriched in each signature**

Hypergeometric enrichment per culture axis (* p<0.05). RRA-744 is enriched for hypoxia/serum/interferon; perturbation signatures for cell-cycle. Culture load alone does not predict transfer (Spearman rho=0.16, ns); the data-driven panel is ~5% canonical osteo-identity genes vs ~92% for curated sets.

